



10 – Estimators and MLE

Tutor: Chin Yi

Recap: Bias, Standard Error & MLE

$$\text{Bias}(\hat{\theta}) = \mathbb{E}[\hat{\theta}] - \theta.$$

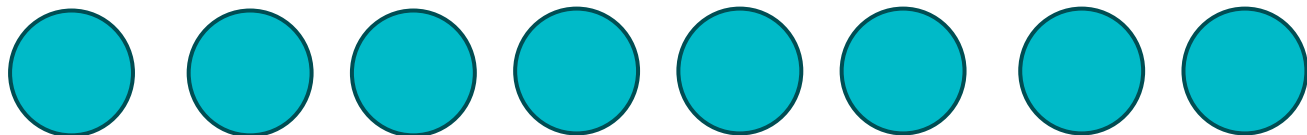
Bias = 0 if it's **unbiased**.

$$SE(\hat{\theta}) = \sqrt{\text{Var}[\hat{\theta}]}.$$

$$L(X_1 = \mathbf{x}_1, \dots, X_n = \mathbf{x}_n | \theta) = \prod_{i=1}^n f(\mathbf{x}_i | \theta)$$

Recap: MLE Intuition

Box with
 p blue balls
 $100-p$ green balls



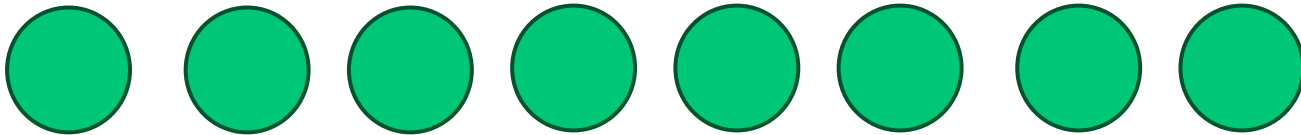
What is an educated guess for the number of p blue balls?

Maybe pick p that is the most probable to get this situation?

$P = 100$

Recap: MLE Intuition

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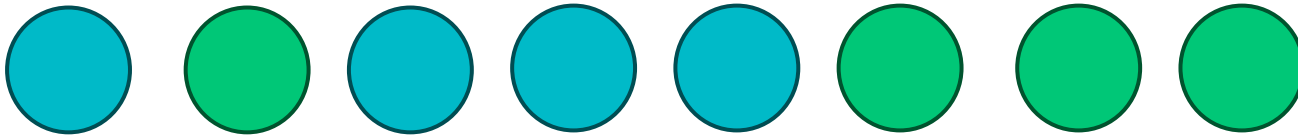
What is an educated guess for the number of p blue balls?

Maybe pick p that is the most probable to get this situation?

$P = 0$

Recap: MLE Intuition

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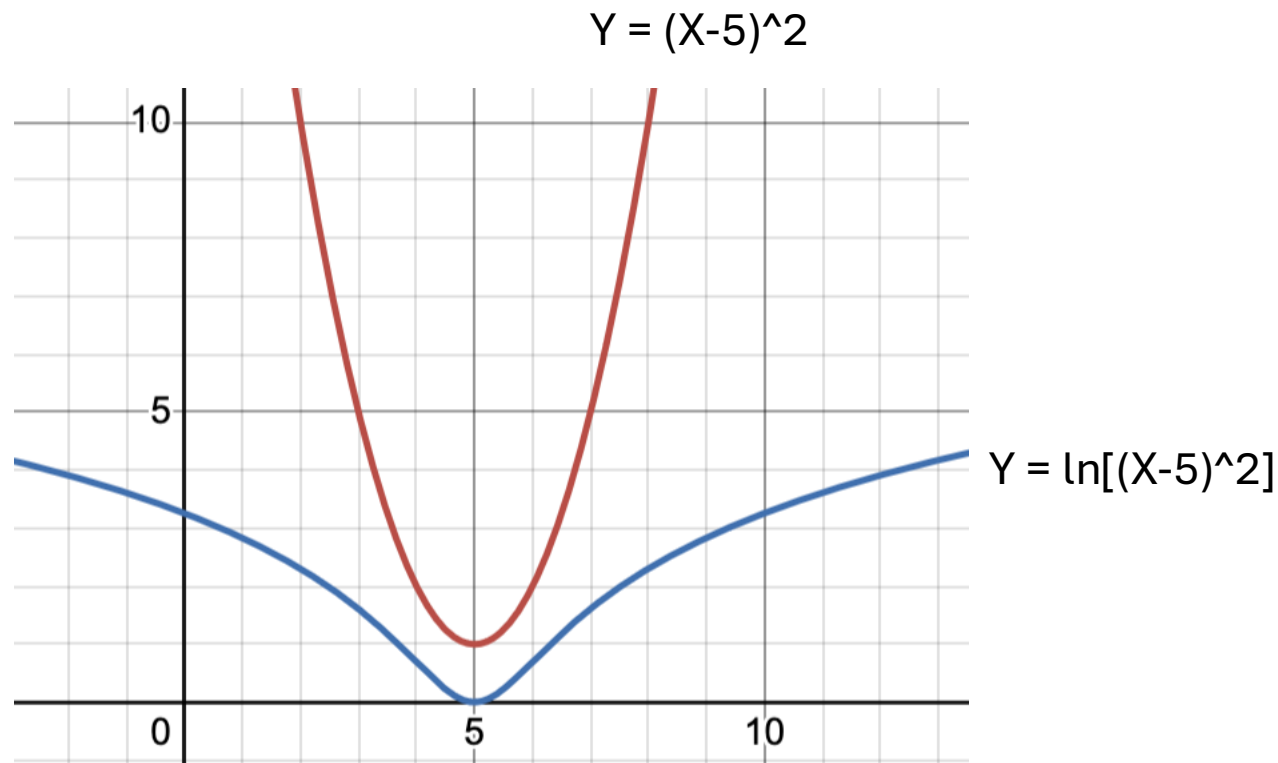
What is an educated guess for the number of **p** blue balls?

Maybe pick **p** that is the most probable to get this situation?

$$L(X_1 = x_1, \dots, X_n = x_n | \theta) = \prod_{i=1}^n f(x_i | \theta)$$

P = 50

Recap: MLE use of Ln



If the equation is too **complicated**, maybe the use of **ln** can help to simply the equation.

Ultimately, the estimator we need to find is the same.

Usually if f is an exponential, use $\ln$.

Problem 1 (Bias and Standard Error of Parameter Estimators)

Let D_θ , $0 \leq \theta \leq 1$, be the discrete distribution with the following PMF:

x	0	1	2	3
$f(x)$	$2\theta/3$	$\theta/3$	$2(1-\theta)/3$	$(1-\theta)/3$

and $f(x) = 0$ otherwise. Let $X_1, \dots, X_n$ be an i.i.d random sample drawn from D_θ and let $\bar{X}$ denote the sample mean. We consider the following estimators for θ .

$$\begin{aligned}\hat{\theta}_1(n) &= -\frac{1}{2}\bar{X} \\ \hat{\theta}_2(n) &= \frac{7 - (X_1 + X_2 + X_3)}{6} \\ \hat{\theta}_3(n) &= \frac{7 - 3\bar{X}}{6} \\ \hat{\theta}_4(n) &= \frac{1}{16} \left(17 - \frac{3}{n} \sum_{i=1}^n X_i^2 \right)\end{aligned}$$

(a) Which of the these estimators are unbiased?

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$$\hat{\theta}_1(n) = -\frac{1}{2}\bar{X}$$

$$\hat{\theta}_2(n) = \frac{7 - (X_1 + X_2 + X_3)}{6}$$

$$\hat{\theta}_3(n) = \frac{7 - 3\bar{X}}{6}$$

$$\hat{\theta}_4(n) = \frac{1}{16} \left(17 - \frac{3}{n} \sum_{i=1}^n X_i^2 \right)$$

$$\text{Var}(\hat{\theta}_1(n)) = (-1/2)^2 \text{Var}(\bar{X}) = \frac{1}{4n} \left(-4\theta^2 + 4\theta + \frac{2}{9} \right) = \frac{1}{n} \left(-\theta^2 + \theta + \frac{1}{18} \right),$$

$$\text{Var}(\hat{\theta}_2(n)) = (-1/6)^2 (\text{Var}(X_1) + \text{Var}(X_2) + \text{Var}(X_3)) = -\frac{1}{3}\theta^2 + \frac{1}{3}\theta + \frac{1}{54},$$

$$\text{Var}(\hat{\theta}_3(n)) = (-3/6)^2 \text{Var}(\bar{X}) = \frac{1}{4n} \left(-4\theta^2 + 4\theta + \frac{2}{9} \right) = \frac{1}{n} \left(-\theta^2 + \theta + \frac{1}{18} \right),$$

$$\text{Var}(\hat{\theta}_4(n)) = \left(\frac{3}{16n} \right)^2 \sum_{i=1}^n \text{Var}(X_i^2) = \frac{9}{256n} \left(-\frac{256}{9}\theta^2 + \frac{208}{9}\theta + \frac{50}{9} \right) = \frac{1}{n} \left(-\theta^2 + \frac{13}{16}\theta + \frac{25}{128} \right).$$

(b) For each of these estimators, compute the standard error.

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(b) For each of these estimators, compute the standard error.

$$\begin{aligned}SE(\hat{\theta}_1(n)) &= \sqrt{\frac{1}{n} \left(-\theta^2 + \theta + \frac{1}{18} \right)}, \\ SE(\hat{\theta}_2(n)) &= \sqrt{-\frac{1}{3}\theta^2 + \frac{1}{3}\theta + \frac{1}{54}}, \\ SE(\hat{\theta}_3(n)) &= \sqrt{\frac{1}{n} \left(-\theta^2 + \theta + \frac{1}{18} \right)}, \\ SE(\hat{\theta}_4(n)) &= \sqrt{\frac{1}{n} \left(-\theta^2 + \frac{13}{16}\theta + \frac{25}{128} \right)}.\end{aligned}$$

Problem 1 (Bias and Standard Error of Parameter Estimators)

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- For $\hat{\theta}_3(n)$: An estimation of θ is

$$\hat{\theta}_3 = (7 - 3\bar{x})/6 = (7 - 3 \times 1.5)/6 = 5/12,$$

here $\bar{x} = 1.5$ is obtained from the given sample.

- For $\hat{\theta}_4(n)$: An estimation for θ is

$$\hat{\theta}_4 = \frac{1}{16} \left(17 - \frac{3}{10} \sum_{i=1}^{10} x_i^2 \right) = \frac{71}{160}.$$

(c) The following observations for $X_1, \dots, X_n$ are given (here $n = 10$):

3, 0, 2, 1, 3, 2, 1, 0, 2, 1

For each *unbiased* estimator from above, substitute the observations into the estimator to obtain an estimation for θ .

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and $f(x) = 0$ otherwise. Let $X_1, \dots, X_n$ be an i.i.d random sample drawn from D_θ and let $\bar{X}$ denote the sample mean. We consider the following estimators for θ .

(d) If the unknown parameter θ occurs in the formula for the standard error $SE(\hat{\theta})$, we can replace θ by $\hat{\theta}$ to get an *estimated* standard error, denoted by $\widehat{SE}(\hat{\theta})$. For each estimator found in part (c), compute its etimated standard error.

$$SE(\hat{\theta}_2(n)) = \sqrt{-\frac{1}{3}\theta^2 + \frac{1}{3}\theta + \frac{1}{54}},$$
$$SE(\hat{\theta}_3(n)) = \sqrt{\frac{1}{n} \left(-\theta^2 + \theta + \frac{1}{18} \right)},$$
$$SE(\hat{\theta}_4(n)) = \sqrt{\frac{1}{n} \left(-\theta^2 + \frac{13}{16}\theta + \frac{25}{128} \right)}.$$

(d)

- For $\hat{\theta}_2(n)$: The estimated standard error is

$$\widehat{SE}(\hat{\theta}_2(n)) = \sqrt{-\frac{1}{3}\hat{\theta}_2^2 + \frac{1}{3}\hat{\theta}_2 + \frac{1}{54}} \approx 0.304.$$

- For $\hat{\theta}_3(n)$: The estimated standard error is

$$\widehat{SE}(\hat{\theta}_3(n)) = \sqrt{\frac{1}{10} \left(-\hat{\theta}_3^2 + \hat{\theta}_3 + \frac{1}{18} \right)} \approx 0.173.$$

- For $\hat{\theta}_4(n)$: The estimated standard error is

$$\widehat{SE}(\hat{\theta}_4(n)) = \sqrt{\frac{1}{10} \left(-\hat{\theta}_4^2 + \frac{13}{16}\hat{\theta}_4 + \frac{25}{128} \right)} \approx 0.19.$$

Problem 2 (Bias and Standard Error)

Let $X_1, X_2, \dots, X_n$ be i.i.d (random sample) from the exponential distribution whose PDF is $f(x|\theta) = \frac{1}{\theta}e^{-x/\theta}$, where $x > 0, \theta > 0$.

(a) Show that $\bar{X}$ is an unbiased estimator of θ .

(b) Show that the variance of $\bar{X}$ is $\frac{\theta^2}{n}$.

Mean | Variance

$\text{Expo}(\theta), \theta > 0$		$\frac{1}{\theta}e^{-x/\theta}, 0 \leq x < \infty$		$\begin{cases} 1 - e^{-x/\theta} & \text{if } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$		$\theta = \frac{1}{\lambda}$		$\theta^2 = \frac{1}{\lambda^2}$
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Problem 3 (Maximum Likelihood Estimation)

Let $X_1, \dots, X_n$ be an i.i.d with PDF

$$f(x|\theta) = \frac{\theta}{\sqrt{2\pi}} e^{-\frac{\theta^2 x^2}{2}} \text{ for all } x \in \mathbb{R},$$

where $\theta \in (0, \infty)$ is an unknown parameter. Compute the MLE for θ based on the observations

$$x_1 = 1.5, x_2 = 2.2, x_3 = 1.3, x_4 = 3.5, x_5 = 3.3.$$

From the observations, we have

$$\sum_{i=1}^5 x_i^2 = 1.5^2 + 2.2^2 + 1.3^2 + 3.5^2 + 3.3^2 = 31.92.$$

Hence the MLE for θ is

$$\theta = \sqrt{\frac{5}{31.92}} \approx 0.396.$$

Geometric(p)	$(1 - p)^{x-1}p, \quad x \geq 1$	$\frac{1}{p}$	$\frac{1-p}{p^2}$
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Problem 4 (Maximum Likelihood Estimation)

Let $X_1, \dots, X_n$ be an i.i.d from the geometric distribution $Geom(p)$, where $0 < p < 1$ is an unknown parameter. Compute the MLE for p based on the observations

$$x_1 = 2, x_2 = 3, x = 4$$

Problem 5 (Maximum Likelihood Estimation and Bias)

Let $X_1, \dots, X_n$ be i.i.d from the distribution with PDF $f(x|\theta) = \frac{1}{\theta}x^{(1-\theta)/\theta}$, $0 < x < 1$, $\theta > 0$. Show that the maximum likelihood estimator of θ is

$$-\frac{1}{n} \sum_{i=1}^n \ln X_i.$$

SFT – T3



https://go.blueja.io/kdIQL__F4UuYCvcLzOMOEQ

SFT – T4



https://go.blueja.io/YA_TcNtgs0un0H6PLESZpw